

1. Scope of Work

Development of a location-based gaming platform consisting of:

- Mobile Applications (Android & iOS): Authentication, QR/manual code game access, offline game engine, GPS tasks, multimedia questions, gamification, multiplayer, leaderboards.
- Web Administration Platform: Game creation studio, rules engine, session management, analytics & reporting.
- Backend Infrastructure: Node.js + PostgreSQL + Redis (scalable), secure APIs, VPS/cloud-ready.
- User Experience: Accessible, offline-first, multilingual, gamification-driven.

2. Project Roadmap & Timeline

We propose a structured, milestone-based delivery approach, ensuring transparency, predictability, and alignment throughout the development lifecycle. The timeline below outlines the phased execution plan:

Weeks 1–2: Kickoff & Foundation Setup

- Conduct kickoff meetings, finalize scope, success KPIs (mapped to performance metrics).
- Define system architecture: services, database schema, GPS/geo-design, multiplayer strategy (peer vs. server authoritative).
- Set up DevOps: CI/CD pipelines, multi-region infrastructure, backups, monitoring & observability.
- Deliverables: Architecture document, deployment playbook, CI pipelines.

Weeks 3–5: Core Backend & Mobile (MVP)

- Backend: API skeleton, authentication (OAuth/JWT + refresh tokens), user/profile management, basic endpoints.
- Mobile app: UI scaffolding, onboarding flow, core navigation, basic gameplay client integrated with engine.
- Game engine v0: rules engine, turn logic, asset pipeline.
- Testing: Unit + basic integration tests, automated builds.
- Deliverables: Staging backend, Alpha APK/IPA, basic game creation UI.

Weeks 5–7: Game Creation Tools & Alpha Testing

- Develop simple game creator UI (scenario/template creation).
- Enable save/load templates.
- Conduct internal alpha testing (bugs, usability, balance checks).
- Stabilization & alpha feedback documentation.
- Deliverables: Alpha release, test report, prioritized backlog.

Weeks 7–9: Offline Functionality + Advanced Content Types

- Implement offline mode with local storage & sync engine.
- Add advanced interactions (timed/multimedia questions, conditional branching).
- Edge-case testing for roaming & sync conflicts.
- Deliverables: Offline-capable builds, test logs & documentation.

Weeks 9–12: Multiplayer & Admin Platform

- Multiplayer backend: matchmaking, sockets/WebRTC, in-game sync.
- Core multiplayer flows: lobby, invite system, state sync.
- Web-based admin platform: user management, publishing, analytics hooks.
- Scale testing on staging.
- Deliverables: Multiplayer MVP, Admin platform v1, load test report.

Weeks 12–15: Analytics & Reporting

- Integrate event-tracking (Mixpanel/GA + custom events).
- Build analytics pipelines, dashboards & enterprise reporting endpoints.
- Deliverables: Analytics dashboards, reporting API, documentation.

Weeks 15–18: Security Hardening & Compliance

- Pen-testing, vulnerability scans, remediation.
- Encryption-at-rest/in-transit, RBAC, auditing.

- GDPR/other compliance checklist.
 - Production cutover planning.
 - Deliverables: Security report, compliance checklist, production playbook.
- Weeks 18–20: Production Launch (Go-Live)
- Soft/staged rollout with KPI monitoring (crash, load times, GPS).
 - Triage support & apply hotfixes.
 - Release marketing & store launch.
 - Deliverables: Public release, release notes, monitoring dashboards.

3. Technology Stack

Layer Technology Purpose

Frontend Next.js (latest), React,

TailwindCSS

SEO-friendly SSR/SSG, fast

UI, responsive

Analytics Google Analytics 4 (GA4) Page views & booking CTA tracking

SEO Meta tags, sitemap, robots.txt

Discoverability & search ranking

Comms WhatsApp Deep Links Instant booking flow

Backend Node.js, PostgreSQL, Redis APIs, scalable data handling, caching

Mobile Apps Android & iOS (React Native)

Game client, GPS tasks, offline-first play

Admin Panel Web-based Admin (React, Game creation studio, Node.js) rules, analytics

Infrastructure VPS / Cloud (CI/CD, monitoring, backups)

Secure deployment, scalability, observability

4. Project Milestone & Payment Schedule

Milestone Timeline

Milestone 1 Week 1–2 Kickoff & Foundation Setup

Milestone 2 Week 3–4 Core Backend & Mobile (MVP)

Milestone 3 Week 5–6 Game Creation Tools & Alpha Testing

Milestone 4 Week 7–8 Offline Functionality +

Advanced Content Types

Milestone 5 Week 9–10 Multiplayer & Admin Platform

Milestone 6 Week 11–12 Analytics & Reporting

Milestone 7 Week 13–14 Security Hardening & Compliance

Milestone 8 Week 15–16 Production Launch (Go-Live)

